

Documentation Map

Date: 2026-04-26

Purpose

This file is a quick guide to the documentation in this repository. It is meant to answer three questions:

- what each document is about;
- whether it is current, duplicated, or mostly historical;
- where to start depending on the kind of context needed.

How to read this map

Status labels used here:

- **current**: useful as an active reference for the present analysis workflow
- **current-duplicate**: same content exists elsewhere in Markdown or another PDF
- **historical**: useful for analysis history, early design choices, or debugging context
- **plot-review**: mostly plot dumps, validation comments, or discussion material

Best starting points

If you only want the most useful current documentation, start here:

1. `project_status.md`
2. `documentation/NORMALIZATION_DOCUMENTATION.md`
3. `documentation/presentation_mutau_weights.md`
4. `documentation/slides_combine.md`
5. `MuTau_channel/slides/slides_tmva_combine.md`
6. `MuTau_channel/slides/slides_weights.md`

If you want the broader analysis history, start here:

1. `documentation/05_02_25.pdf`
2. `documentation/07_08.pdf`
3. `documentation/16_10.pdf`
4. `updates/25_02_02.pdf`
5. `documentation/plots.pdf`

Repository-wide summaries

README.md

Status: **current**

Use for:

- quick MuTau-oriented checkpoint overview
- proton-requirement plots
- event-count comparisons
- short notes on proton-variable handling

Notes:

- focused on a specific analysis checkpoint rather than the whole repository
- useful operationally, but not a full technical note

README.pdf

Status: `current-duplicate`

Use for:

- PDF version of the README content

Notes:

- no clear added value over the Markdown version

project__status.md

Status: `current`

Use for:

- high-level overview of the three channels
- phase-0 / phase-1 / proton-tagging structure
- current open issues and next steps

project__status.pdf

Status: `current-duplicate`

Use for:

- PDF export of the project status note

Current MuTau documentation

documentation/NORMALIZATION_DOCUMENTATION.md

Status: `current`

Use for:

- MC normalization formula
- generator-weight sums
- DY and ttbar normalization constants
- relation between phase-1 event weights and final plotting weights

documentation/NORMALIZATION_DOCUMENTATION.pdf

Status: `current-duplicate`

Use for:

- PDF export of the normalization note

documentation/presentation_meeting.md

Status: **current**

Use for:

- short status note on the proton-mixing bug
- acceptance measurement discussion
- weight-structure summary

documentation/presentation_meeting.pdf

Status: **current-duplicate**

documentation/presentation_mutau_weights.md

Status: **current**

Use for:

- phase-0 / phase-1 / proton-mixing workflow
- event weight definitions
- proton pool description
- data-driven proton acceptance explanation

documentation/presentation_mutau_weights.pdf

Status: **current-duplicate**

documentation/presentation_mutau_weights_slides.pdf

Status: **current-duplicate**

Use for:

- slide version of the same proton-mixing and weight-normalization story

documentation/presentation_weights.pdf

Status: **current-duplicate**

Use for:

- shorter weight-oriented presentation version

documentation/slides_combine.md

Status: **current**

Use for:

- Combine workflow
- shape production
- datacard structure
- nuisances, rate parameters, and asymptotic limits

documentation/slides_combine_26_03.pdf

Status: current-duplicate

documentation/slides_mutau.md

Status: current

Use for:

- MuTau BDT training/application overview
- weight evolution
- plotting after proton requirements

documentation/slides_mutau.pdf

Status: current-duplicate

documentation/slides_mutau_05_02.pdf

Status: current-duplicate

documentation/slides_mutau_07_03.pdf

Status: current-duplicate

documentation/slides_mutau_26_02.pdf

Status: current-duplicate

MuTau_channel/slides/slides_tmva_combine.md

Status: current

Use for:

- multi-method TMVA summary
- propagation of MVA outputs to Combine
- method-comparison framing

MuTau_channel/slides/slides_weights.md

Status: current

Use for:

- detailed documentation of `sinal.py`
- signal normalization and cutflow
- PPS fiducial logic
- xi systematics and radiation-damage weights

Current MuTau PDF mirrors in MuTau_channel/slides/

These are useful if a PDF is more convenient, but they mostly duplicate Markdown-backed content already present in the repository:

- MuTau_channel/slides/NORMALIZATION_DOCUMENTATION.pdf — current-duplicate
- MuTau_channel/slides/presentation_meeting.pdf — current-duplicate
- MuTau_channel/slides/presentation_mutau_weights.pdf — current-duplicate
- MuTau_channel/slides/presentation_mutau_weights_slides.pdf — current-duplicate
- MuTau_channel/slides/presentation_weights.pdf — current-duplicate
- MuTau_channel/slides/slides_combine_13_04.pdf — current
- MuTau_channel/slides/slides_combine_16_04.pdf — current
- MuTau_channel/slides/slides_combine_26_03.pdf — current-duplicate
- MuTau_channel/slides/slides_mutau.pdf — current-duplicate
- MuTau_channel/slides/slides_mutau_05_02.pdf — current-duplicate
- MuTau_channel/slides/slides_mutau_07_03.pdf — current-duplicate
- MuTau_channel/slides/slides_mutau_13_04.pdf — current
- MuTau_channel/slides/slides_mutau_26_02.pdf — current-duplicate
- MuTau_channel/slides/slides_weights.pdf — current

Notes:

- the dated April PDFs are useful because they can contain the latest slide-state even when the older Markdown note is more generic
- the February and March slide PDFs mostly duplicate content already represented in the Markdown versions

Historical and early-development PDFs

documentation/05_02_25.pdf

Status: `historical`

Use for:

- original physics motivation
- anomalous tau magnetic moment framing
- explanation of why exclusive $\gamma \gamma \rightarrow \tau \tau$ is interesting
- early list of decay channels

Best value:

- shows the broad scientific motivation before the repository settled into its current MuTau-centered workflow

documentation/first-stage-analysis.pdf

Status: `historical`

Use for:

- long early-development slide notebook
- first signal variable studies
- early correlation studies

- evolution of the project presentation

Best value:

- useful for reconstructing the earliest signal-oriented thinking and variable choices

documentation/07_08.pdf

Status: `historical`

Use for:

- early codeflow and dataflow
- older phase-0 / phase-1 file organization
- old background normalization assumptions
- bug notes from proton enrichment

Best value:

- one of the best docs for understanding the older TauTau-oriented workflow and why some legacy code exists

documentation/16_10.pdf

Status: `historical`

Use for:

- transition from the old proton-handling workflow to the newer one
- what changed in processing for data/QCD vs DY/ttbar
- summary of signal-processing logic and proton cuts

Best value:

- best historical explanation of the move toward preserving proton variables in real data products

documentation/27_10.pdf

Status: `historical`

Use for:

- later-stage plot review
- signal and data plot presentation
- BDT input visualization

Best value:

- useful mainly as a snapshot of what was being visually checked at that stage

documentation/01_08_2025.pdf

Status: `historical`

Use for:

- early weekly-meeting style slides

Notes:

- overlaps with the project-motivation material found in `first-stage-analysis.pdf`

documentation/06__11 (1).pdf

Status: `historical`

Use for:

- later development checkpoint

Notes:

- should be consulted if you want to reconstruct week-to-week project evolution

documentation/12__09.pdf

Status: `historical`

documentation/21__08.pdf

Status: `historical`

documentation/28__08.pdf

Status: `historical`

documentation/Cópia de apresentacao.pdf

Status: `historical`

documentation/apresentacao.pdf

Status: `historical`

Notes for the last five:

- these appear to be intermediate presentation decks rather than enduring technical references
- they are useful if you need timeline context, but not the best place to learn the current workflow

Plot-review and debugging material

documentation/plots.pdf

Status: `plot-review`

Use for:

- variable and correlation plots
- old weighting assumptions
- control-region plot observations
- comments about inconsistencies in histogram ranges and shapes

Best value:

- good for understanding what plot-level issues were being debugged

updates/25_02_02.pdf

Status: **plot-review**

Use for:

- short progress update
- event-count changes after fixes
- early signal/background plot comparisons
- summary of variable studies

documentation/29_10-questions regarding plots (1).pdf

Status: **plot-review**

Use for:

- questions and concerns raised about plotted results

Notes:

- useful for internal review context rather than for the nominal analysis workflow

Recommended usage by purpose

If you want to understand the current analysis

Read:

1. project_status.md
2. documentation/presentation_mtau_weights.md
3. documentation/NORMALIZATION_DOCUMENTATION.md
4. documentation/slides_combine.md
5. MuTau_channel/slides/slides_tmva_combine.md
6. MuTau_channel/slides/slides_weights.md

If you want the original physics motivation

Read:

1. documentation/05_02_25.pdf
2. documentation/first-stage-analysis.pdf

If you want the evolution of proton handling and weighting

Read:

1. documentation/07_08.pdf
2. documentation/16_10.pdf
3. documentation/presentation_meeting.md
4. documentation/NORMALIZATION_DOCUMENTATION.md

If you want plot-validation and control-region context

Read:

1. documentation/plots.pdf
2. updates/25_02_02.pdf
3. documentation/29_10-questions regarding plots (1).pdf

Bottom line

The repository documentation is concentrated around the MuTau workflow, and the Markdown notes are the best active references. The extra PDFs are still valuable, but mostly for:

- project history;
- older weighting/proton-handling conventions;
- early signal-variable studies;
- plot-validation and debugging context.

For everyday analysis work, prefer the Markdown-backed notes first and use the older PDFs as historical supplements.